

Herald



Herald — sherald Flavor Guide (Operator)

Field	Value
Revision	1
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Status	active
Status summary	Nano-detail operator reference for sherald (System Herald) — the host-safety flavor. Documents serve (the /v1/safety_state HTTP plane), the §9/§11.4/§12.6 pre-flight guard commands (destructive-guard, force-push-gate, mem-budget-watch, backup-snapshot, constitution-pull), and version. ANTI-BLUFF: every claim derived from running the built sherald binary (sherald --help, sherald <sub> --help, sherald version --json) and commons/branding.go. The guard commands NEVER execute the destructive op themselves — they classify + exit non-zero to BLOCK; documented as such below.
Issues	(none specific to this guide)
Continuation	Bump when additional §43 sysops bindings land or /v1/safety_state gains fields.

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§1. What sherald is

sherald is **System Herald** — flavor key s, prefix SHR, default serving port **24793**. Per commons/branding.go its mission is "Host-level system events fan-out + safety-state

guards". It is the flavor a host-management wrapper consults before any destructive operation: each guard subcommand is a **pre-flight gate** that classifies the requested op through the HRD-020 constitution binding and exits non-zero to BLOCK it. Per §12/§107 host-safety, sherald never performs the destructive op itself — it only advises the operator's wrapper.

Build it:

```
go build -o /tmp/sherald ./sherald/cmd/sherald
```

§2. The subcommand surface

Verbatim from sherald --help:

Subcommand	What it does
serve	Start the System Herald HTTP server (/v1/safety_state)
destructive-guard	Pre-flight gate for rm/git reset --hard/git push --force (§9.1)
force-push-gate	Pre-flight gate for force-push: merge-first + per-session auth (§11.4.41 / §9.2)
mem-budget-watch	Sample host memory + enforce the §12.6 60% ceiling (one-shot or --watch daemon)
backup-snapshot	Create a hardlinked snapshot before a destructive op (§9.3)
constitution-pull	Wrap fetch + rebase + post-pull validation gate (§11.4.26 + §11.4.32)
version	Print System Herald version + build info
completion	Generate shell autocompletion (Cobra built-in)

§3. version

```
$ sherald version --json
{"arch":"arm64","binary":"sherald","build_time":"unknown","commit":"unknown-dev"}
```

§4. serve — the safety-state HTTP plane

sherald serve starts the System Herald HTTP server. Per docs/INTEGRATION.md §10 it exposes GET /v1/safety_state, returning process-local counters (open events, mem%, last destructive-op log). It shares the commons/cli/serve scaffold with the other serving flavors; the default bind port is 24793.

```
sherald serve --http-port 24793
```

§5. The host-safety guard commands

Each of these is a **gate**: it inspects the requested operation, builds the relevant §-Subject, classifies it through the HRD-020 binding, and **exits non-zero on a FAIL verdict** so the operator's wrapper aborts. None of them mutate host state beyond the explicit, non-destructive action named. Every guard accepts --emit to additionally drive the verdict as a real constitution event.

destructive-guard <op>...

Detects a destructive operation (`rm / git reset --hard / git clean / git push --force`) from the supplied args and whether a preceding hardlinked backup exists, then classifies + exits non-zero on FAIL (BLOCKING the op). It **never executes the op itself**.

Flag	Meaning
<code>--backup-exists</code>	Assert a preceding hardlinked backup exists
<code>--backup-path string</code>	Path to a hardlinked backup snapshot (existence $\Rightarrow$ backup satisfied)
<code>--repo string</code>	Repo dir scoping a relative <code>--backup-path</code>

```
sherald destructive-guard rm -rf ./build --backup-path ./
build.bak-20260530 || echo "BLOCKED"
```

force-push-gate <ref>

Classifies merged-first (`behind upstream == 0`) AND per-session-authorized through the §11.4.41 binding; exits non-zero on FAIL. Never performs the force-push.

Flag	Meaning
<code>--authorized</code>	Assert explicit per-session force-push authorization (or <code>HERALD_FORCE_PUSH_AUTHORIZED</code>)
<code>--upstream string</code>	Upstream ref to prove merge-first against (default <code>origin/main</code>)
<code>--fetch</code>	Fetch before computing merge-first (default: use already-fetched state)
<code>--repo string</code>	Repo dir (default: discovered from CWD)

mem-budget-watch

Samples host memory used-fraction READ-ONLY (`vm_stat/sysctl` on darwin, `/proc/meminfo` on linux — never allocates to test the ceiling), classifies against the §12.6 60% ceiling. One-shot by default (`sample  $\rightarrow$  classify  $\rightarrow$  exit non-zero on breach`); `--watch` loops emitting on breach-transitions until SIGINT.

Flag	Meaning
<code>--watch</code>	Daemon mode: loop sampling, emit on breach-transition until SIGINT
<code>--interval duration</code>	Sample interval in <code>--watch</code> mode (default 10s)

There is also a hidden `--used-fraction` override (or `HERALD_MEM_FRACTION` env) — the deterministic test seam to drive PASS/FAIL branches without real memory pressure.

```
sherald mem-budget-watch # one-shot
sherald mem-budget-watch --watch --interval 30s
```

backup-snapshot <target>

SAFE CREATE: hardlinks a snapshot of <target> (file or dir) at --dest (default <target>.bak-<UTC-timestamp> alongside the target) — cheap and space-free, the §9.3 "hardlinked backup before destructive op" helper. It **never deletes or overwrites** the target. Exits non-zero if the snapshot could not be created.

Flag	Meaning
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--dest string	Snapshot destination (default <target>.bak-<UTC-timestamp>)
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constitution-pull

Pulls the discovered constitution submodule (fetch + rebase against --remote) and runs the §11.4.32 post-pull validation gate, classifying both §11.4.26 (pull ok?) and §11.4.32 (validation passed?). Prefers the canonical constitution_pull.sh / find_constitution.sh when discoverable.

Flag	Meaning
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--remote string	Remote to fetch + rebase against (default origin)
--constitution-dir string	Constitution submodule dir (default: discovered)
--repo string	Repo dir to start discovery from (default: CWD)
--skip-validate	Do not run the post-pull validation gate (records validated=false)
--assume-validated	Treat the post-pull validation gate as PASSED (test seam / operator override)

§6. References

- Source: sherald/cmd/sherald/main.go + sysops_cmds.go.
- Branding: commons/branding.go (flavors).
- Integration: docs/INTEGRATION.md §10 (GET /v1/safety_state).
- Constitution bindings: HRD-020 §9.1 / §9.2 / §9.3 / §11.4.26 / §11.4.32 / §11.4.41 / §12.6.
- Companion flavor guides: docs/guides/{PHERALD, CHERALD, BHERALD, RHERALD, IHERALD, SCHERALD, QAHERALD}.md.

Sources verified

Verified 2026-05-30: internal doc — no external online sources. Every subcommand, flag, default, and gate-behaviour claim above was derived by running the built sherald binary (sherald --help; sherald serve|destructive-guard|force-push-gate|mem-budget-watch|backup-snapshot|constitution-pull --help; sherald version --json) and reading commons/branding.go on 2026-05-30. No flags were invented. Re-verify whenever the sherald Cobra surface changes.